Universidad Corporativa | Programas Académicos

Ingeniería en Desarrollo de Software

academia global

Actividad #2 Calculando direcciones.

Administración de Redes

Y Servidores.

Ingeniería en Desarrollo de Software



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FECHA: 09/11/2026.

Índice

Contextualizacion…… 2

Desarrollo……..…… 3

- Comandos utilizados…….
- Capturas de pantalla……...

Referencias……...

Contextualizacion

Actividad 2 – Calculando direcciones.

Escenario

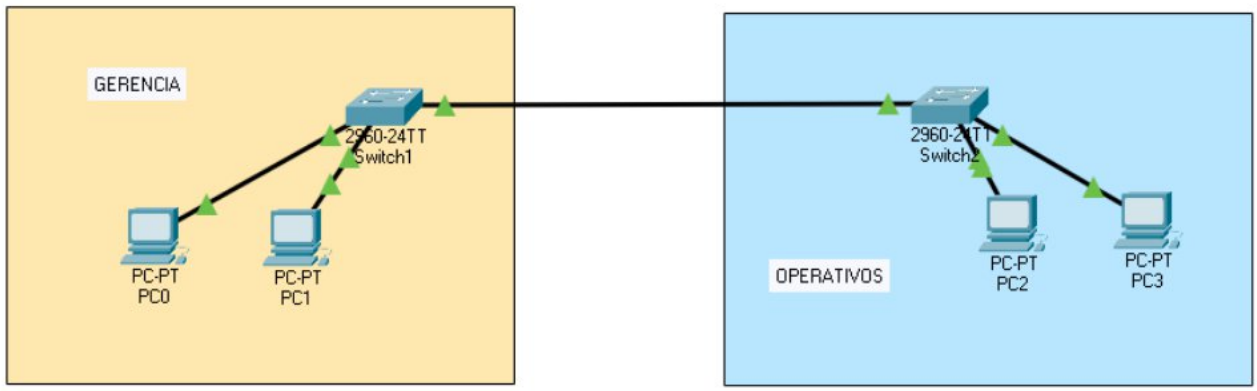
Etapa 2- Cálculo de direcciones

Contextualización:

Rogelio está aplicando para el puesto de administrador de sistemas. Por tanto, se le solicita realizar una propuesta gráfica y simulada para dos redes locales.

Actividad:

Con base a la actividad pasada, calcular las direcciones del direccionamiento de red de la VLAN de gerencia (192.168.10.0) y de operativos (192.168.20.0) y en base a lo obtenido asignar una dirección IP a cada PC.



Desarrollo y capturas de pantalla.

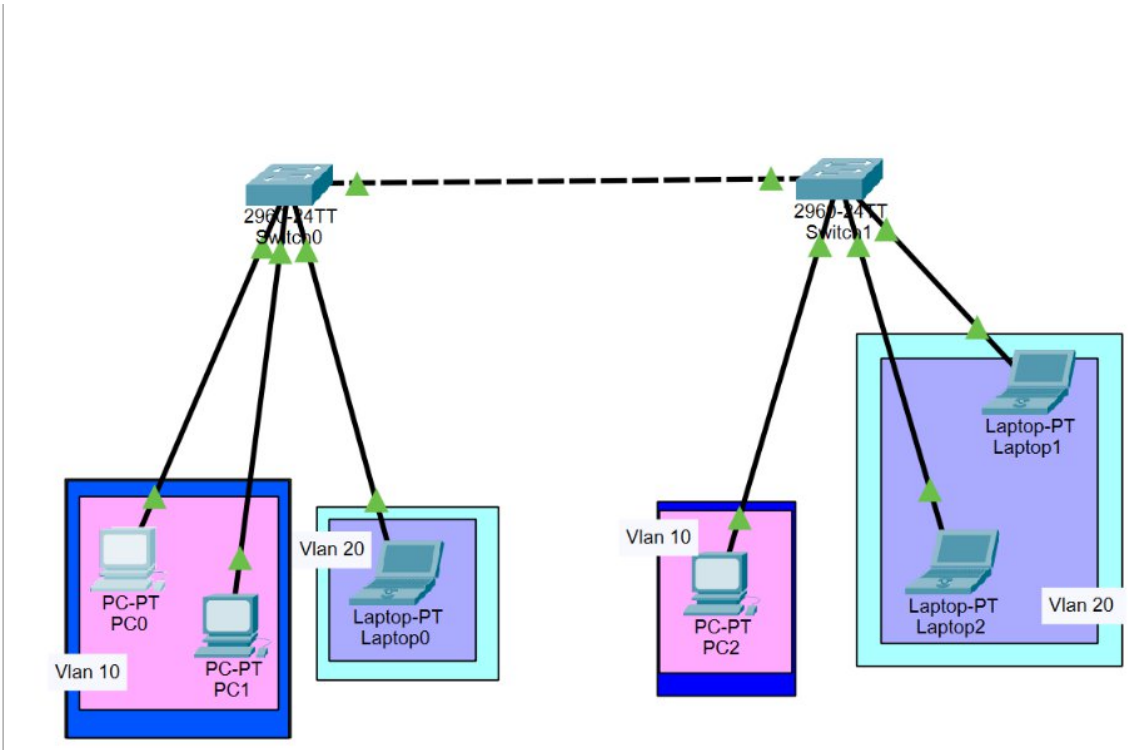
En mi segunda actividad estaré documentando el procedimiento técnico realizado durante lo largo de dicha actividad en la cual seguiremos con el escenario que realizamos en la actividad pasada.

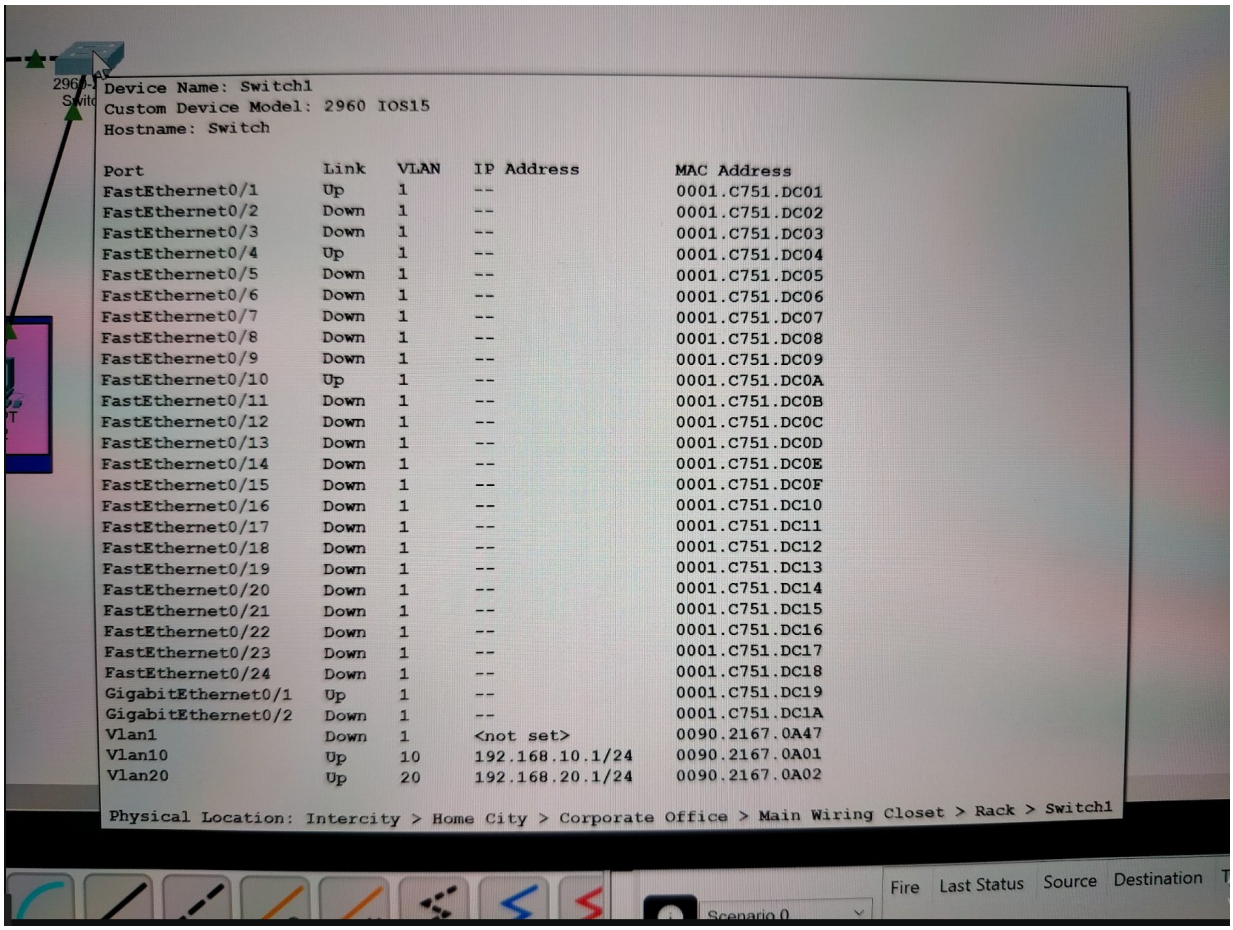
Gracias a esta actividad aprendí a diferenciar claramente la dirección que identifica a la red completa.

Entendí porque es importante tener redes separadas para distintas áreas de una empresa ya que ayuda a organizar el tráfico y evitar saturación en la red.

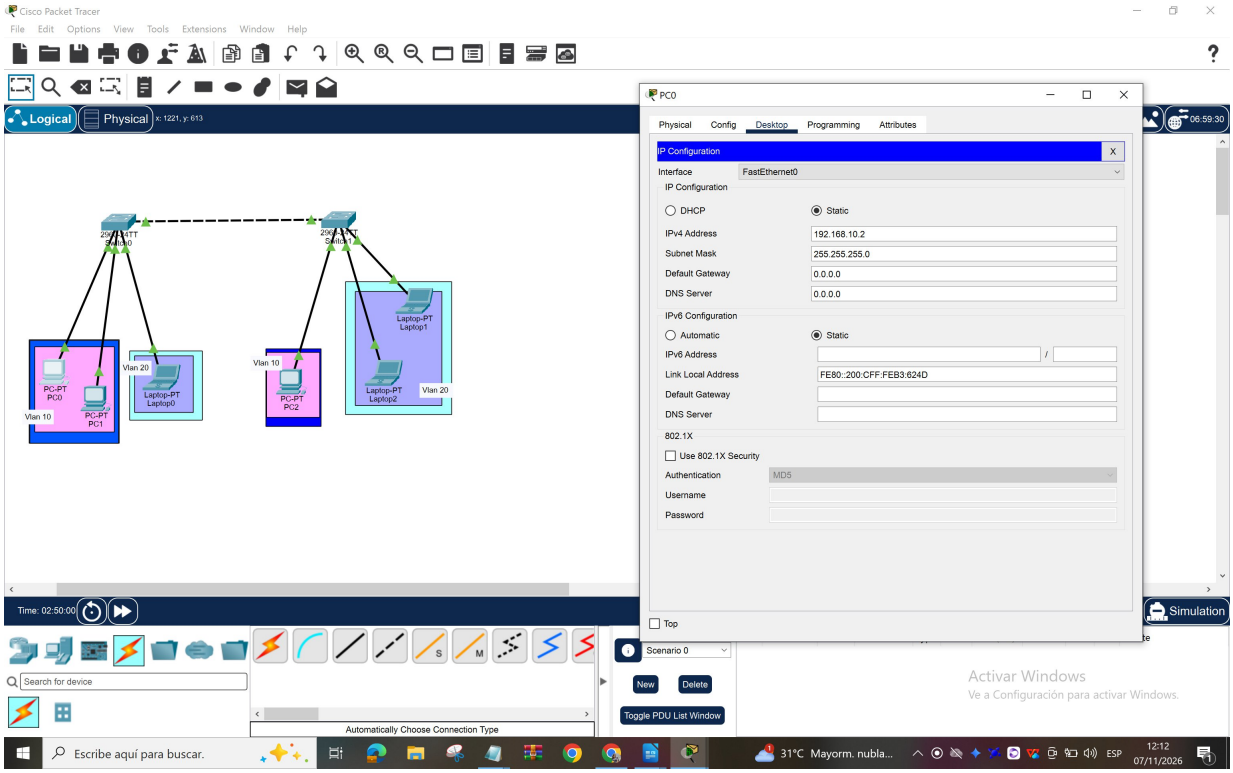
Estuve practicando el uso del comando ipconfig en la consola para checar las configuraciones de los equipos y verificar los parámetros de red rápida.

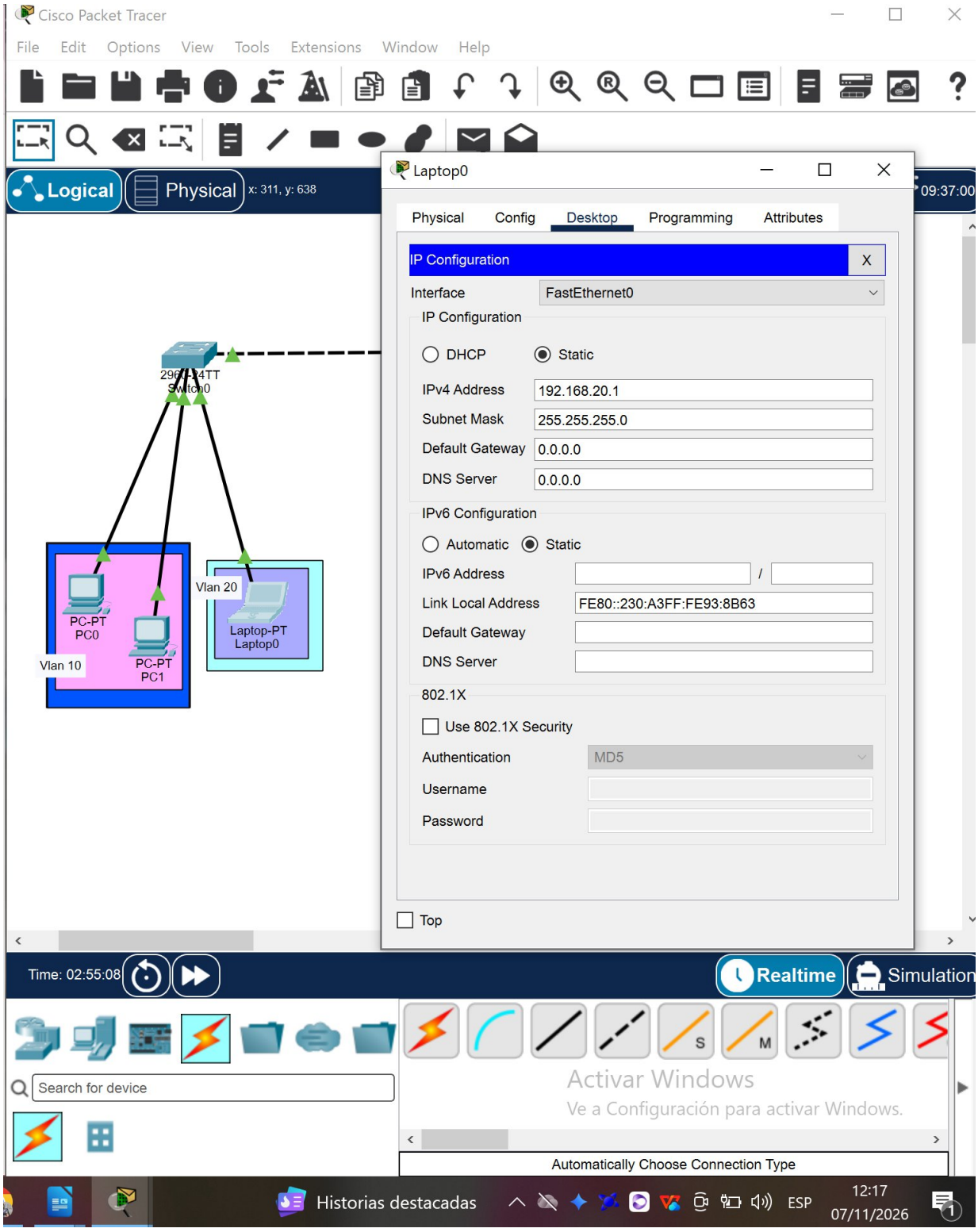
Reforcé el uso del simulador para conectar switches, computadoras y asi configurar parámetros de red estáticos de forma clara y ordenada.

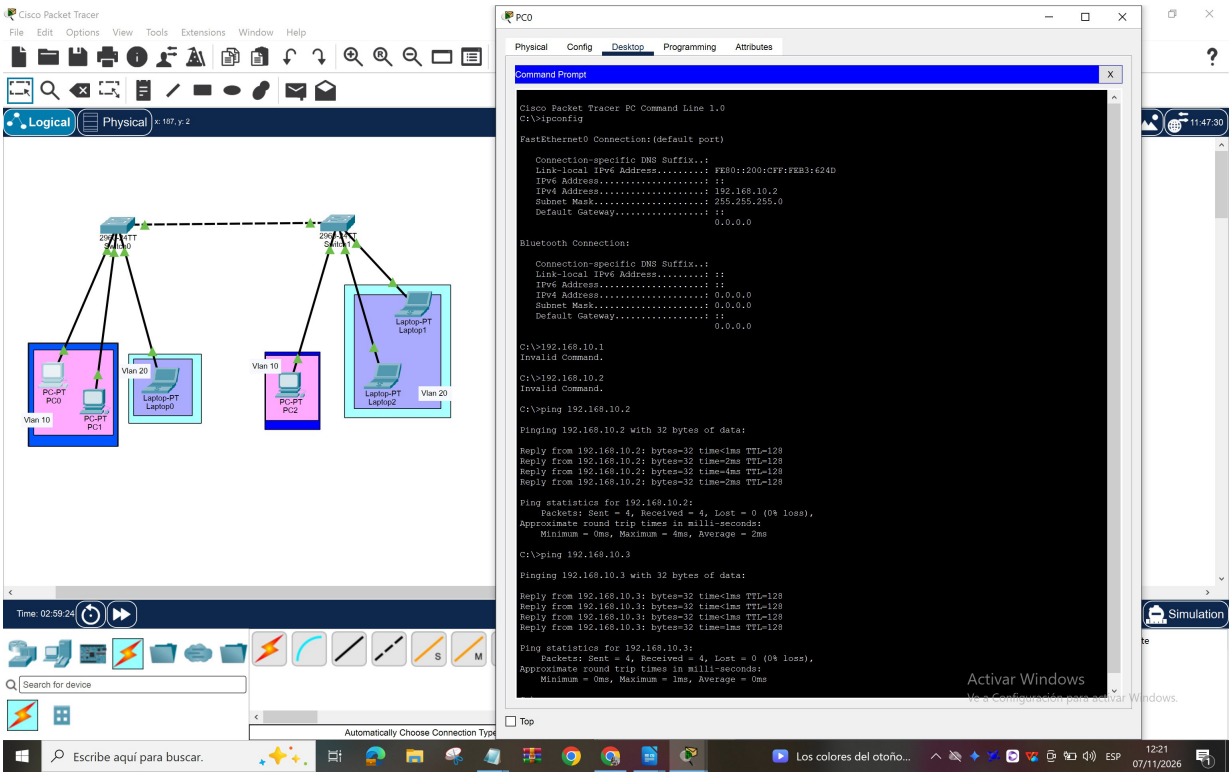




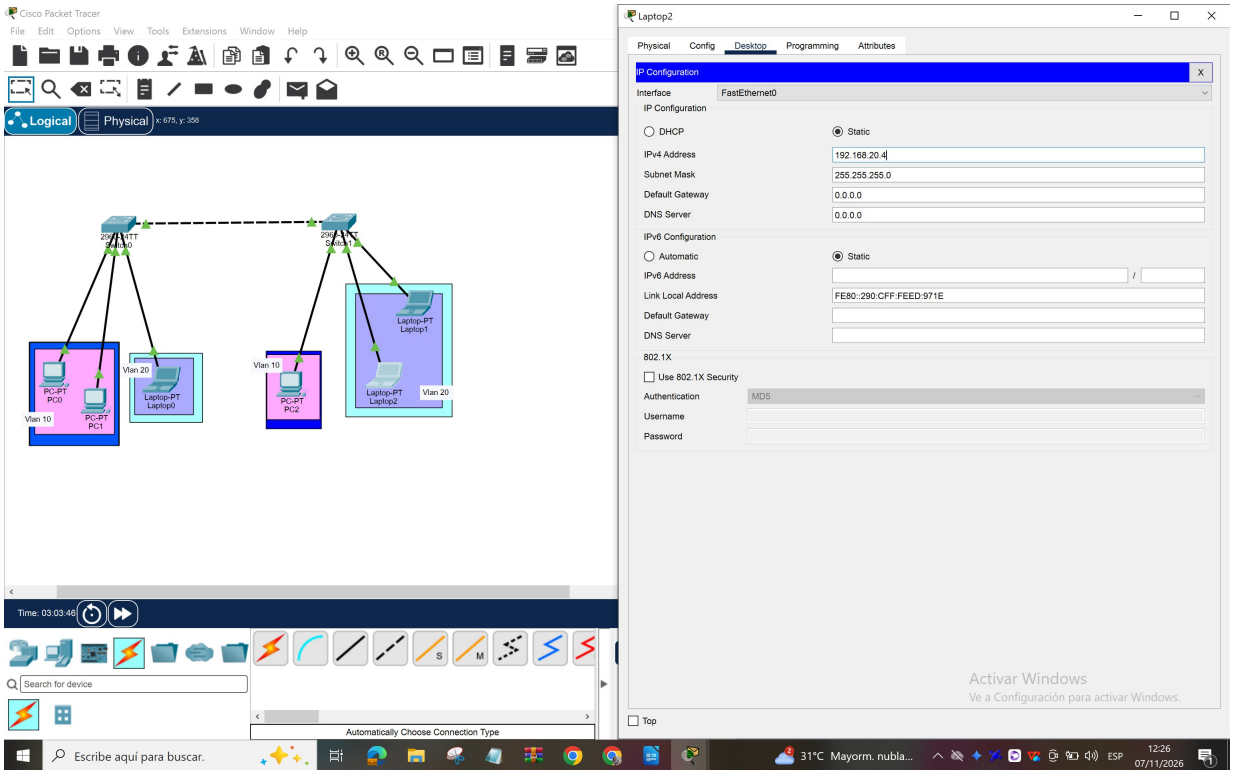
Posteriormente iremos agregando la IP correspondiente a cada computadora.

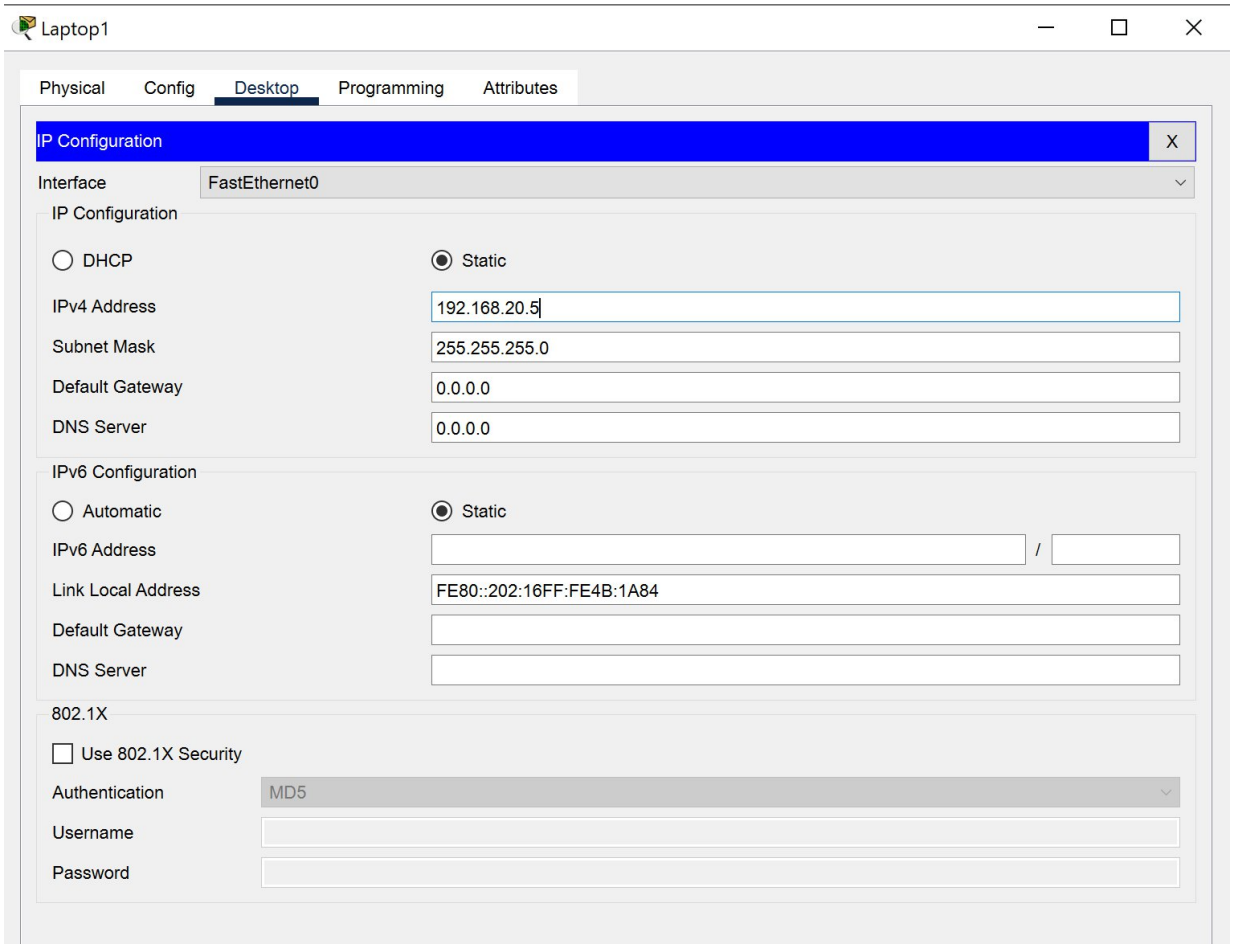


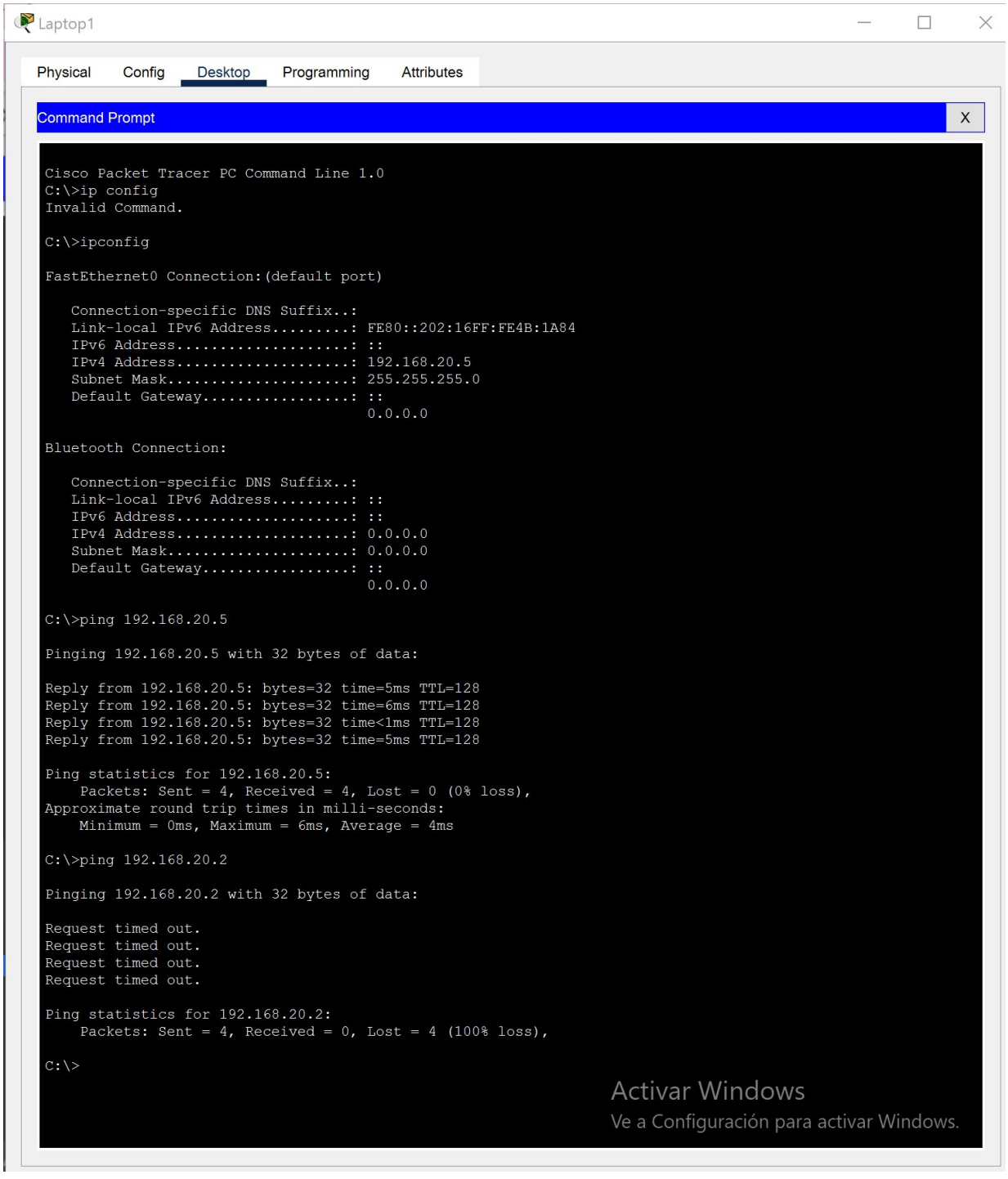

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 311, y: 638
09:37:00
Laptop0
Physical Config Desktop Programming Attributes
IP Configuration
X
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.20.1
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::230:A3FF:FE93:8B63
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Vlan 10
Vlan 20
PC-PT PC0
PC-PT PC1
Laptop-PT Laptop0
Time: 02:55:08
Realtime
Simulation
Search for device
Activar Windows
Ve a Configuración para activar Windows.
Automatically Choose Connection Type
Historias destacadas
ESP
12:17
07/11/2026

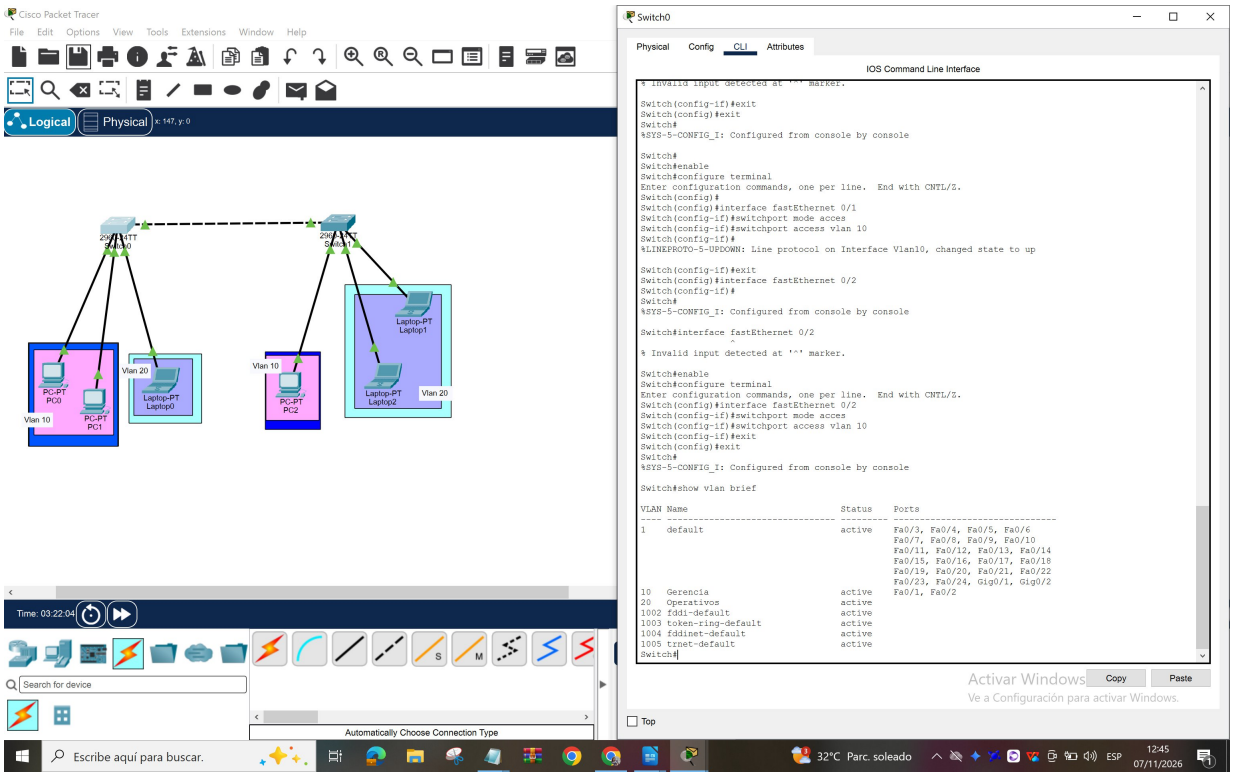


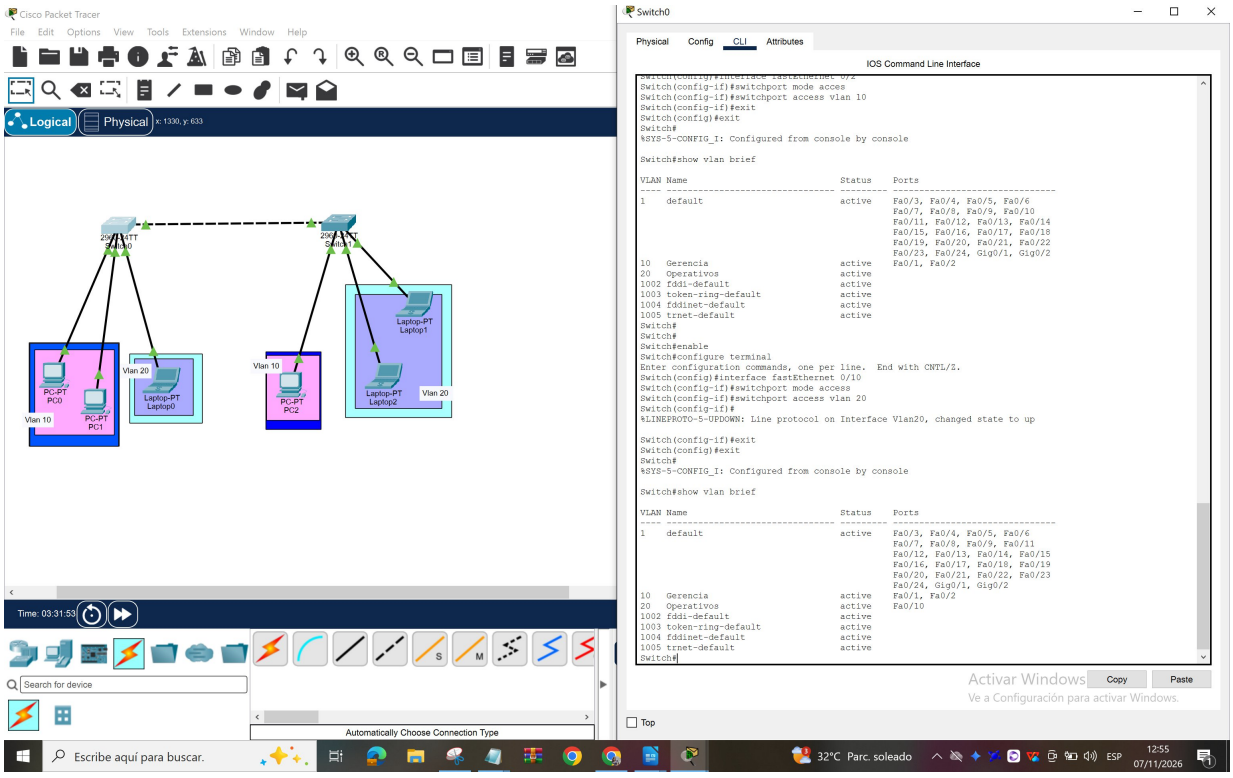
Posteriormente se realizan pruebas.

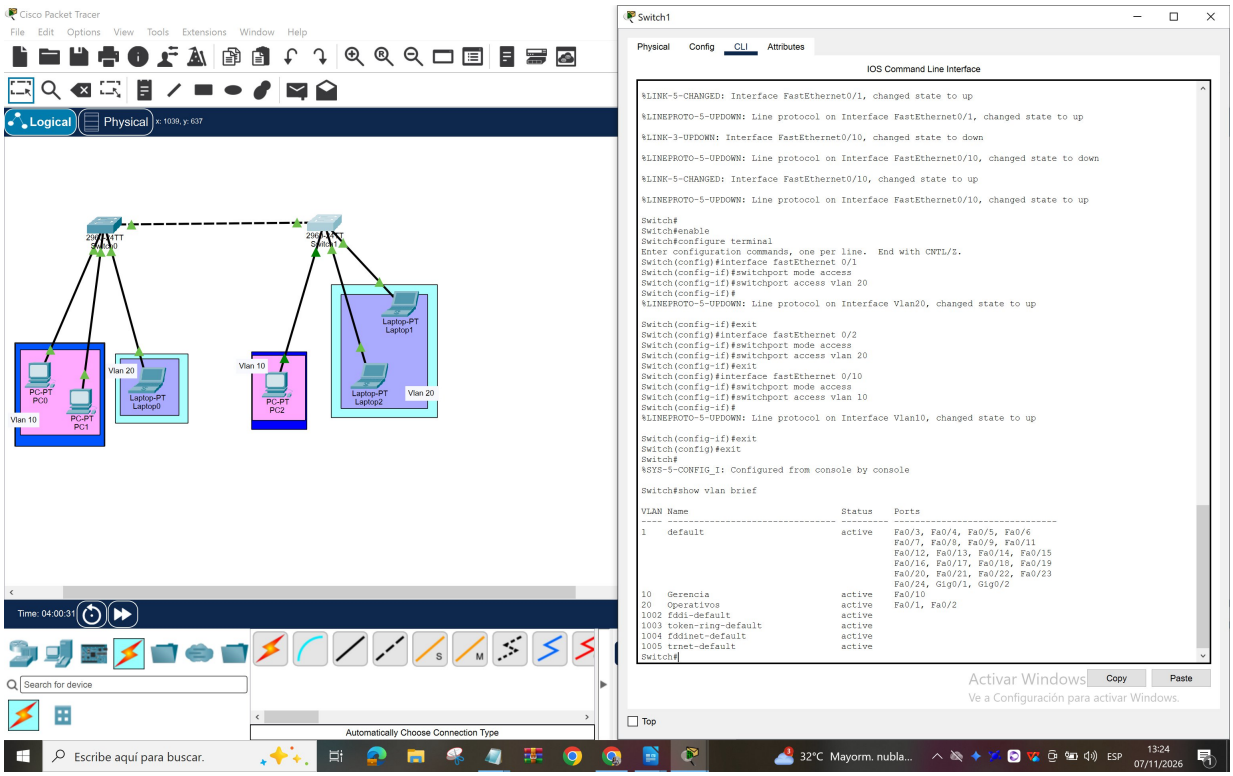



Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.5
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::202:16FF:FE4B:1A84
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ip config
Invalid Command.
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::202:16FF:FE4B:1A84
IPv6 Address.....: ::
IPv4 Address.....: 192.168.20.5
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.20.5
Pinging 192.168.20.5 with 32 bytes of data:
Reply from 192.168.20.5: bytes=32 time=5ms TTL=128
Reply from 192.168.20.5: bytes=32 time=6ms TTL=128
Reply from 192.168.20.5: bytes=32 time<1ms TTL=128
Reply from 192.168.20.5: bytes=32 time=5ms TTL=128
Ping statistics for 192.168.20.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 6ms, Average = 4ms
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Activar Windows
Ve a Configuración para activar Windows.

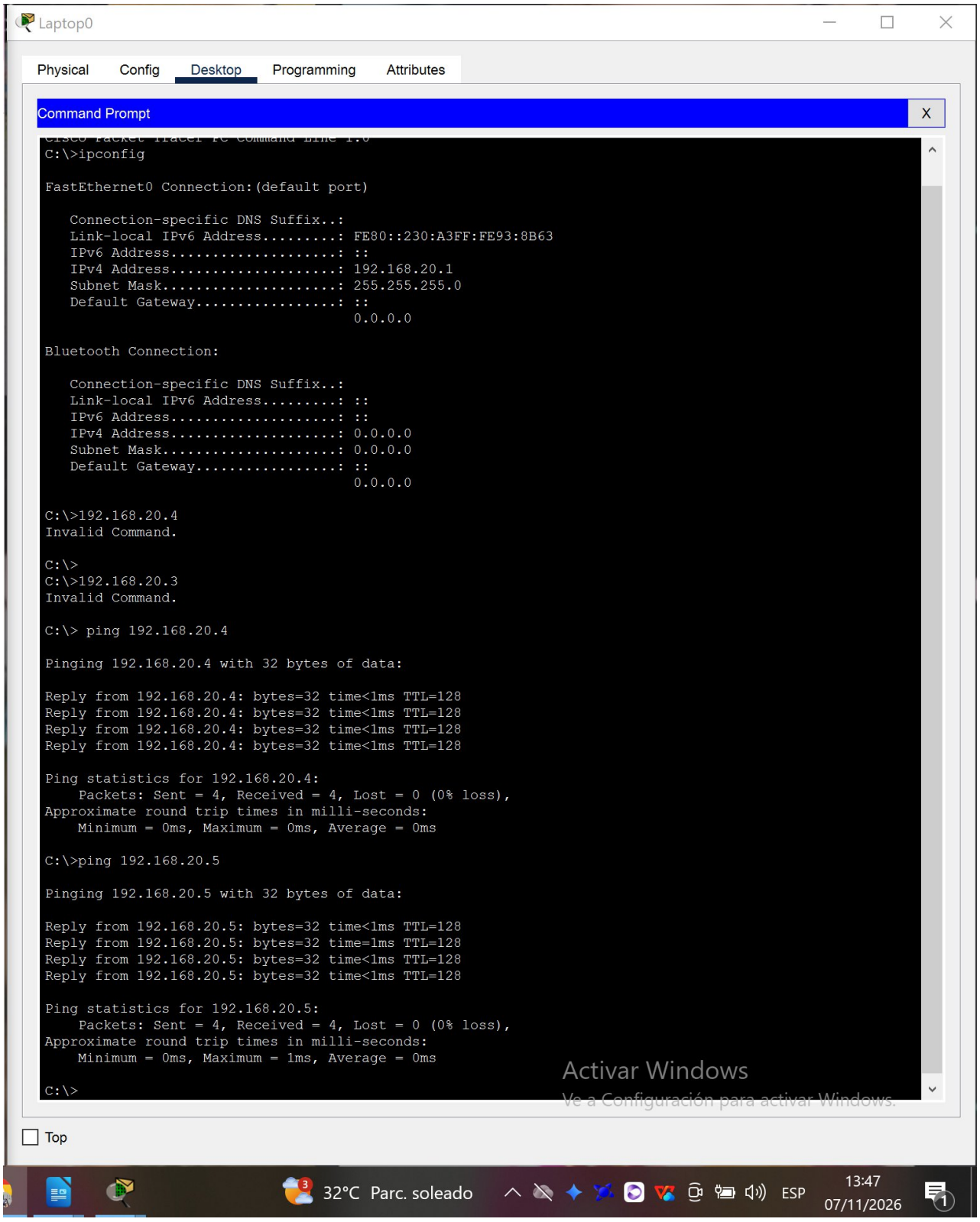

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 147, y: 0
PC-PT PC0
PC-PT PC1
Laptop-PT Laptop0
PC-PT PC2
Laptop-PT Laptop1
Laptop-PT Laptop2
Vlan 10
Vlan 20
Time: 03:22:04
Search for device
Automatically Choose Connection Type
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode acces
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#interface fastEthernet 0/2
% Invalid input detected at '^' marker.
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode acces
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 Gerencia active Fa0/1, Fa0/2
20 Operativos active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Activar Windows
Ve a Configuración para activar Windows.
Copy Paste
Top
Escribe aquí para buscar.
32°C Parc. soleado
ESP
12:45
07/11/2026


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1330, y: 633
PC-PT PC0
PC-PT PC1
Laptop-PT Laptop0
PC-PT PC2
Laptop-PT Laptop1
Laptop-PT Laptop2
Vlan 10
Vlan 20
Time: 03:31:53
Search for device
Automatically Choose Connection Type
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch(config-if)#switchport mode acces
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 Gerencia active Fa0/1, Fa0/2
20 Operativos active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1, Gig0/2
10 Gerencia active Fa0/1, Fa0/2
20 Operativos active Fa0/10
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Activar Windows
Ve a Configuración para activar Windows.
Copy Paste
Top
Escribe aquí para buscar.
32°C Parc. soleado
ESP
12:55
07/11/2026


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Laptop-PT Laptop1
Laptop-PT Laptop2
Laptop-PT Laptop0
PC-PT PC0
PC-PT PC1
PC-PT PC2
Vlan 10
Vlan 20
Time: 04:00:31
Search for device
Automatically Choose Connection Type
Escribe aquí para buscar.
32°C Mayorm. nubla...
ESP
13:24
07/11/2026
Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1, Gig0/2
10 Gerencia active Fa0/10
20 Operativos active Fa0/1, Fa0/2
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Copy
Paste
Top
Activar Windows
Ve a Configuración para activar Windows.



Puerta de enlace



Referencias

Cisco Packet Tracer. (s. f.). https://www.netacad.com/cisco-packet-tracer

iOSMart Academy by Marco Alonso. (2022, 20 octubre). ***Configurar 2 VLAN en dos switches de una red LAN (Parte 1)*** **[Vídeo]. YouTube. https://www.youtube.com/watch?v=iKfZlPxrm-zY**

iOSMart Academy by Marco Alonso. (2022b, octubre 20). ***Configurar 2 VLAN en dos switches de una red LAN utilizando Cisco Packet Tracer Parte 2*** **[Vídeo]. YouTube. https://www.youtube.com/watch?v=7A9AM-sDSDo**

Video conferencing, web conferencing, webinars, screen sharing**. (s. f.). Zoom. https://academiaglobal-mx.zoom.us/rec/play/oYd24e_KintHefDyRTipG_U6uNpJV2A26gELMROkeLFJ-ZIQ5y-tQzt6peL24moe66xbyTN0JyFwKfxLY.qhVjrf7lWvspb4aU?accessLevel=meeting&canPlayFromShare=true&from=share_recording_detail&continueMode=true&oldStyle=true&componentName=rec-play&originRequestUrl=https%3A%2F%2Facademiaglobal-mx.zoom.us%2Frec%2Fshare%2FGgCym_DAVPPjHfxacmXPufCRGnMZVSW7IHGV-xWze0ix6Y2h3E6xx3P9LzIO81YHj.bMBnJMu6b9xl2A7w**

GitHub - VERONICANIKEY/ACT.-1-ESCENARIO-LAN-: PERIODO 6. MATERIA: **(s. f.). GitHub. https://github.com/VERONICANIKEY/ACT.-1-ESCENARIO-LAN-**